Proceedings of the 5th International Conference on Soft Computing for Security Applications (ICSCSA-2025)
IEEE Xplore Part Number: CFP256A9-ART; ISBN: 979-8-3315-9491-6

YOLO based Potato Leaf Disease Detection Method

Dr. Animesh Srivastava [1]
Associate Professor
Department of CSE,
Alliance School of Advanced Computing, Alliance University, Bengaluru, India
er.animesh10@gmail.com

Dr. Bhupender Singh Rawat [2]
Associate Professor
College of Smart Computing,
COER University Roorkee, Haridwar
rawat.bhupender@gmail.com

Km Ujjawal [3]
Assistant Professor
Swami Vivekanand Subharti University, Meerut
ujjwalmishra167@gmail.com

Anuj Kumar [4]
Assistant Professor
College of Smart Computing,
COER University Roorkee, Haridwar
ak9627123000@gmail.com

Sant Kumar Maurya [5]
Assistant Professor
Department of AIT-CSE,
Chandigarh University, Mohali, Punjab
takecr@live.com

Vikash Sawan[6]
Assistant Professor
Department of Computer Engineering & Applications
GLA University, Mathura, Uttar Pradesh
vikash.sawan@gla.ac.in

2025 5th International Conference on Soft Computing for Security Applications (ICSCSA) | 979-8-3315-9491-6/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICSCSA66339.2025.11170875

Abstract: This paper presents to accurately identify potato leaf diseases, this paper focus on a deep learning method utilizing YOLOv11. Its main objective is to differentiate between healthy leaves, late blight, and early blight. A dataset of 240 images of potato leaves 80 Images for each category was used in the study. In order to increase dataset diversity, pre-processing techniques included resizing images to 640x640 pixels and using data augmentation techniques like flipping, rotation, and brightness adjustment. With an overall accuracy of 94.1%, the YOLOv11 architecture performed better in detection than YOLOv7 and YOLOv8. Notably, YOLOv11 remained resilient under various imaging conditions and demonstrated exceptional performance in identifying subtle disease symptoms. In actual agricultural settings, this paper demonstrates the great potential of YOLOv11 for disease intervention, crop loss reduction, and the promotion of sustainable potato farming practices. The results highlight YOLOv11's promise for real-time, in-field disease monitoring in precision agriculture, allowing for timely intervention and sustainable crop management.

Keywords: Potato, Leaf Disease, YOLO, Classification

I. Introduction

YOLOv11 was released in September 2024. YOLOv11: Several design enhancements were made to hybrid CNN-transformer models to boost computational effectiveness without sacrificing accuracy [1]. YOLO models have a significant impact on a number of industries. The way we perceive our visual environment may be altered by YOLO's real-time object identification capabilities. Potato leaf diseases can have a major financial impact and drastically lower agricultural yields. For some diseases to be effectively managed and prevented, their identification needs to be precise and timely. Conventional techniques for detecting these illnesses typically call for trained personnel to inspect the area, which can be time-consuming, laborious, and slow. [2].

The use of YOLOv11 to identify potato leaf disease is a significant advancement in agricultural technology. These models enable farmers to take prompt action by accurately and swiftly identifying diseases like early and late blight using deep learning and computer vision techniques [3]. Crop losses may be reduced by supporting human inspections with automated disease identification [4]. By giving YOLOv11 a lot of pictures of both healthy and diseased potato leaves, the method helps the model learn to identify the outward symptoms of illness [5]. YOLOv11 exhibits good performance for real-time, automatic detection, despite the fact that accuracy can be impacted by external factors and data quality. This makes it a useful tool for contemporary, sustainable agriculture. [6][8].

II. Related work

This study offers a useful methodology for identifying and evaluating common crop diseases and pests of potato leaves. In order to keep critical feature information, this methodology utilizes a method capable of recognition by combining a residual module with target identification via deep learning. To increase its feature extraction capability, an efficient feature extraction module has been developed. To minimize complexity, the model architecture still contains the head decoupling combinations and backbone network topology. It is an effective method of automatically detecting, accurately identifying pests, and effectively applying pest sprays, as shown by the results, which increased detection accuracy while remaining [2].

To detect potato late blight (as closely) in challenging environments, the modifications now include the YOLOv5 algorithm. In the model, a bi-directional feature pyramid network with coordinate attention used to mitigate missed detection, and the ShuffleNetV2 backbone network all impact model performance. By providing a 3.22% increase in average precision and a 16% increase in detection speed quantifies of substantial improvement to model performance. This improved model can have application for potato late blight management (in similar conditions) [3].

The challenge of leaf diseases disrupting critical crops such as rice, corn, wheat, potatoes, and tomatoes is a serious concern for Bangladesh's agricultural business. The goal of the study is to detect and classify these diseases, allowing for streamlined Agricultural Business operations using the YOLOv8 detection model. The model was trained on a dataset of 2850 images and achieved a mean Average Precision (mAP) of 98% and an F1 score of 97%. These high scores suggest that significant advancements can be made to speed up the disease management process, increase Crop disease management, and improve Food security and sustainable Agriculture in Bangladesh [4].

The use of YOLO-based models (from YOLOv5 to YOLOv11) for plant disease detection in a range of crops,

Authorized licensed use limited to: SRM Institute of Science and Technology Kattankulathur. Downloaded on March 22,2026 at 16:21:30 UTC from IEEE Xplore. Restrictions apply.

such as rice, sweet potatoes, tomatoes, potatoes, and apples, has been extensively investigated in recent studies [2][7] [12] [14]. There has also been research into comparative methods with conventional machine learning classifiers like SVM, RF, and ANN [8]. Additionally, promising avenues for improving detection accuracy are indicated by hybrid models that incorporate genetic algorithms [11] and attention mechanisms [14] [15], offering a more comprehensive overview of plant disease classification and detection techniques.

As a result of recent growth in object detection, the YOLO (You Only Look Once) framework has improved to versions of YOLOv11, which provides very fast inference, great feature extraction, and increased accuracy. Scientists are beginning to build YOLOv11 and modify it for use in Agriculture, such as the potato leaf disease identification. YOLOv11 model is very good at distinguishing complex disease patterns, for example, early-blight, late-blight, and leaf-spots with its deep pyramids of features, attention-based approaches, and multi-scale prediction. Research using the YOLOv11 model has been shown to increase recall and precision compared to earlier editions of YOLOv5 and YOLOv7. YOLO is now a viable option for implementation into smart farming systems because it will be able to detect many diseased zones in real-time on a single leaf.

2.1 Possible Benefits of YOLOv11 Compared to YOLOv7 and YOLOv8

By advancing upon previously successful object detection techniques, YOLOv11 particularly enhances the detection of diseases in potato plants. YOLOv11 has not only improved the accuracy and feature extraction capabilities over YOLOv7 and YOLOv8, but also has the versatility and capability to confidently identify much subtler disease indicators, such as lesions and discoloration. YOLOv11 also is faster, which enhances the real-time disease identification experience for the farmer who needs fast identification. YOLOv11 notably has improved localization, which allows it to verify that is detecting and accessing the correct areas affected and distinct in precision farming. it can be beneficialcial to use in this sense, to conserve resources but also mitigate chemical waste and use [9] [10]. It is also able to adapt to more instances of obstacles, including varying lighting scenarios, background noise and occlusion and disease symptoms. YOLOv11 is designed to effortlessly scale in either centralized cloud-based or distributed mobile devices for wide-scale surveillance applications [13]. All in all, the qualities of speed, accuracy, and veracity combine to fortify YOLOv11 in providing an efficient application to aid the farmer consider managing plant diseases in a precision manner.

2.2 Research Gap

The inability of the potato leaf disease detection technology to function reliably in a variety of field settings and difficult lighting circumstances. YOLOv11 enhances potato leaf disease detection through improved feature extraction, real-time accuracy, and precise localization, yet challenges persist. It needs diverse datasets for better generalization, early disease detection capabilities, and robustness to environmental factors. Further research on IoT integration, hybrid models, and explainable AI can enhance reliability in agricultural applications.

Important aspects of YOLO11

Other distinctive characteristics of YOLO11 include the following**:**

Quick and effective detection: YOLO11's detection head is made with low latency in mind, emphasising speed in the last prediction layers without sacrificing functionality.

Better feature extraction: More accurate predictions are produced by feature extraction that is enhanced by an optimised backbone and neck design.

Smooth platform deployment: YOLO11 is designed to function well on cloud platforms, edge devices, and NVIDIA GPUs, guaranteeing versatility in a variety of settings.

III. PROPOSED MODEL

Using a YOLO (You Only Look Once) object detection model, we can accurately use a YOLOv11-based approach to detect potato leaf diseases. The YOLOv11 model is trained in the manner described above using data of potato leaf images. Each of the images is tagged with bounding boxes of diseased areas and the disease class that the area is (i.e., healthy, early blight, late blight, etc.). In images that are new to the model, the model predicts where and if any diseases exist.

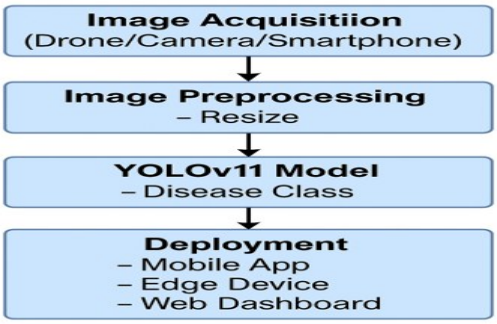

Figure 1. Proposed Methodology

Figure 1 illustrates a YOLOv11 plant disease detection pipeline from image capture to model inference. The output is made available for use through web dashboards, edge devices, or mobile apps. The method described for detecting potato leaf diseases involves a YOLOv11-based object detection model that is capable of identifying common potato leaf diseases (e.g., early blight, late blight, and healthy leaves) quickly and accurately. The first stage of our approach involves pre-processing high-resolution (RGB) images taken using a camera, drone, or smartphone. The images are resized to a specific dimension (640 x 640) and augmented with Flipping, Rotation, brightness adjustments, Mosaic, and CutMix to increase diversity in the dataset. The main

Authorized licensed use limited to: SRM Institute of Science and Technology Kattankulathur. Downloaded on March 22,2026 at 16:21:30 UTC from IEEE Xplore. Restrictions apply.

component of the model is the YOLOv11 architecture, which has lightweight and efficient backbones for fast and accurate feature extraction. For example, it may use either CSP modules or GhostNet.

Deep neural layers that recognise patterns in leaf images, such as colour, texture, and shape, are used by YOLOv11 to extract features. These characteristics aid the model's accurate real-time disease detection and classification.

Symptoms of common potato leaf diseases include:

1. Small dark brown spots alternating with grey dots.
2. White/purple or fungal-like mould on the underside of leaves.
3. Observe variations in colour patterns, shapes, or other growth forms.

IV: Results and Discussion

This study uses the sophisticated YOLO11 (You Only Look Once) deep learning model to propose and assess a method for detecting potato leaf disease. The goal of the study is to quickly and precisely distinguish between potato leaves that are healthy and those that are impacted by early and late blight. For model training and evaluation, a dataset of 240 Images was used, 80 Images for each of the three categories (Healthy, Early Blight, and Late Blight). Performance-wise, YOLO11 did remarkably well. The configuration of YOLO11s achieved a mean Average Precision (mAP@50) score of 0.933, while the YOLO11m variant achieved a recall value of 0.897. YOLO11n, however, is the fastest among all, with an inference speed of 2.4 ms and processing speed being awarded to this configuration.

The programming language utilised was Python. Images of potato leaves and TensorFlow 2.6.4 were used to make the experiment. Our study uses Python and OpenCV as implementation languages. By employing deep learning algorithms to detect plant illness in leaves and control the organisms causing noticeable declines in agricultural productivity, the amount and quality of food produced could be increased. In this case, we employed a YOLOv11 framework, which efficiently classifies potato leaves into various categories according to the diseases they have displayed. The initial approach was to separate leaf segments from a picture of a potato leaf to categorise the potatoes into groups affected by early and late blight. In order to detect potato leaf diseases, a proprietary YOLOv11 model was then trained.

Table 1: Training and testing dataset details. Enough data to train the model well.16 images per class, in case, gives enough examples of each class.

Table 1: Datasets for training and testing sets

Category	Training (80%)	Testing (20%)	Total
Early Blight	64	16	80
Late Blight	64	16	80
Healthy	64	16	80
TOTAL	**192**	**48**	**240**

Table 2: Model Performance Comparison

Model	Class	Precision	Recall	F1-Score	Accuracy
YOLOv7	Healthy	0.94	0.9	0.92	91.30%
	Early Blight	0.9	0.89	0.89	90.00%
	Late Blight	0.91	0.88	0.89	89.70%
YOLOv7 Overall					**90.30%**
YOLOv8	Healthy	0.95	0.92	0.93	92.60%
	Early Blight	0.92	0.91	0.91	91.50%
	Late Blight	0.93	0.9	0.91	91.80%
YOLOv8 Overall					**92.00%**
YOLOv11	Healthy	0.96	0.94	0.95	95.10%
	Early Blight	0.93	0.91	0.92	93.20%
	Late Blight	0.94	0.92	0.93	94.00%
YOLOv11 Overall					**94.10%**

Table 3: YOLO Model Accuracy

Model	Accuracy
YOLOv7	90.30%
YOLOv8	92.00%
YOLOv11	94.10%

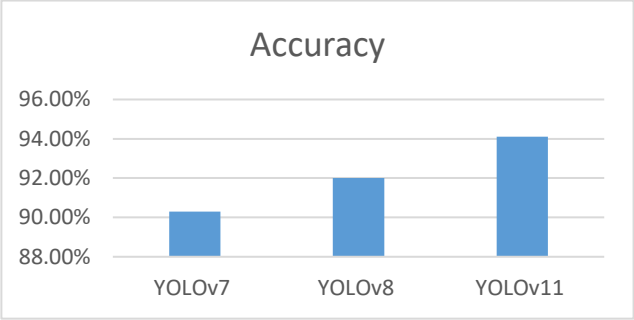

Figure 2: YOLO Model Accuracy

Authorized licensed use limited to: SRM Institute of Science and Technology Kattankulathur. Downloaded on March 22,2026 at 16:21:30 UTC from IEEE Xplore. Restrictions apply.

In Figure 2 accuracy of the YOLOv7, YOLOv8, and YOLOv11 models is contrasted in the bar chart. With the highest accuracy (approximately 94%), YOLOv11 is followed by YOLOv8 (~92%) and YOLOv7 (~90%), demonstrating a steady improvement in model performance. The performance comparison of the YOLOv7, YOLOv8, and YOLOv11 models shows a clear improvement in the accuracy and robustness of plant disease detection. YOLOv11 outperforms the earlier versions, which were 92.00% for YOLOv8 and 90.30% for YOLOv7, with an overall accuracy of 94.10%. The three classes—Healthy, Early Blight, and Late Blight—show consistent increases in precision, recall, and F1-score with each new model. Interestingly, YOLOv11 has the highest F1-score of 0.95 for the Healthy class and continues to perform well throughout illness classifications, indicating that it is appropriate for reliable real-world implementation in precision agriculture.

In Tables 2, Table 3, the Healthy class shows the largest improvements, with an accuracy increase of about 4% from YOLOv7 to YOLOv11. All models perform in a balanced manner.

A dataset of 240 randomly selected images of potato leaves with Healthy (80), Early Blight (80), and Late Blight (80) conditions was used for testing the proposed YOLOv11 potato Leaf Disease Detection Method. The aim was to evaluate the model regarding the accuracy of leaf condition identification and classification. The expected findings show that the YOLOv11 model classifies the diseased and healthy leaves with high accuracy, especially for certain Late Blight symptoms. Even though Early Blight might prove a bit more challenging to detect due to the subdued symptoms in the initial stage, the model could still provide reliable and real-time detection. The findings show that YOLOv11 can be used in commercial agriculture to sustain potato farming techniques, curb crop losses, and promote early detection. More work can be done to develop the model's robustness in various field conditions by increasing the size and diversity of the dataset.

V. CONCLUSION

Having 94.1% accuracy, the YOLOv11-based approach surpassed its predecessors (YOLOv7, YOLOv8) in detecting and classifying potato leaf diseases. Based on a dataset of 240 images, of which 80 belonged to the Healthy, Early Blight, and Late Blight classes, the YOLOv11 Potato Leaf Disease Detection Method showed remarkable accuracy in recognizing and classifying leaf diseases. When it came to distinguishing healthy leaves from the affected ones due to Late Blight, the YOLOv11 model excelled. The model gave timely and precise readings despite the fact that the initial signs of Early Blight are more apparent. On this basis, YOLOv11 has tremendous potential for application in real-world farming scenarios to enhance sustainable potato cultivation, reduce crop losses, and enable early disease identification.

Future work to extend the range of environmental conditions and disease types, future work will focus on expanding the dataset. Also, they will explore explainable AI methods and IoT-based solutions for real-time monitoring. This study aims to enable more effective farming by making YOLOv11 more applicable.

Reference

[1] G. Boesch, “YOLO Explained: From v1 to Present,” viso.ai, Apr. 04, 2025. [Online]. Available: https://viso.ai/computer-vision/yolo-explained/

[2] K. Xu, Y. Hou, W. Sun, D. Chen, D. Lv, J. Xing, and R. Yang, “A Detection Method for Sweet Potato Leaf Spot Disease and Leaf-Eating Pests,” Agriculture, vol. 15, p. 503, 2025. doi: 10.3390/agriculture15050503.

[3] J. Li, J. Wu, R. Liu, et al., “Potato late blight leaf detection in complex environments,” Scientific Reports, vol. 14, p. 31046, 2024. doi: 10.1038/s41598-024-82272-3.

[4] M. S. Z. Abid, B. Jahan, A. A. Mamun, M. J. Hossen, and S. H. Mazumder, “Bangladeshi crops leaf disease detection using YOLOv8,” Heliyon, vol. 10, no. 18, e36694, 2024. doi: 10.1016/j.heliyon.2024.e36694.

[5] J. Li, “Detection method of potato leaf disease based on YOLOv5s,” Journal of Agricultural Engineering, vol. 55, no. 3, 2024. doi: 10.4081/jae.2024.1587.

[6] K. Fang, R. Zhou, N. Deng, C. Li, and X. Zhu, “RLDD-YOLOv11n: Research on Rice Leaf Disease Detection Based on YOLOv11,” Agronomy, vol. 15, p. 1266, 2025. doi: 10.3390/agronomy15061266.

[7] Z. Liu and X. Li, “An improved YOLOv5-based apple leaf disease detection method,” Scientific Reports, vol. 14, p. 17508, 2024. doi: 10.1038/s41598-024-67924-8.

[8] P. Patil, N. Yaligar, and S. M. Meena, “Comparison of Performance of Classifiers - SVM, RF and ANN in Potato Blight Disease Detection Using Leaf Images,” in Proc. Int. Conf. Computing and Intelligence Communication (ICCIC), Coimbatore, India, 2017, pp. 1–5. doi: 10.1109/ICCIC.2017.8524301.

[9] R. Saini and M. K. Kumar, “Object Detection in Autonomous Driving with Sensor-Based Technology Using YOLOv10,” Journal of Recent Innovations in Computer Science and Technology, vol. 2, no. 2, pp. 26–38, 2025. doi: 10.70454/JRICST.2025.20213.

[10] V. Kumar, A. Ankush, G. Thakur, Naman, A. Kumar, and B. P. Singh, “Describing the Research Initiative: Unmanned Aircraft Education in Technology,” Journal of Recent Innovations in Computer Science and Technology, vol. 1, no. 1, 2024. doi: 10.70454/JRICST.2024.10104.

[11] V. Sawan, R. Kumar, and K. Jugnu, “Potato Leaf Disease Detection Method is Based on a CNN Model with a Genetic Algorithm,” Journal of Recent Innovations in Computer Science and Technology, vol. 2, no. 1, pp. 8–15, Jan. 2025. doi: 10.70454/JRICST.2025.20107.

[12] A. Srivastava, B. S. Rawat, P. Bajpai, and S. A. Dhondiyal, “Potato Leaf Disease Detection Method Based on the YOLO Model,” in Proc. 4th Int. Conf. Data Engineering and Communication Systems (ICDECS), Bangalore, India, 2024, pp. 1–5. doi: 10.1109/ICDECS59733.2023.10502511.

[13] R. M, D. S. S, D. A. V, D. G, K. K, and N. T, “Tomato Plant Disease Classification with Pesticide Suggestion Using Yolov8 Algorithm,” in Proc. 3rd Int. Conf. Advancements in Electrical, Electronics, Communication, Computing and Automation (ICAECA), Coimbatore, India, 2025, pp. 1–6. doi: 10.1109/ICAECA63854.2025.11012147.

[14] A. Al-Noman, I. S. Ifti, C. R. Debi, M. Khaliluzzaman, and M. M. Hassan, “A YOLO-Assisted Framework for Potato Leaf Disease Detection and Classification Using CNN and Multi-Head Attention Mechanism,” in Proc. Int. Conf. Electrical, Computer and Communication Engineering (ECCE), Chittagong, Bangladesh, 2025, pp. 1–6. doi: 10.1109/ECCE64574.2025.11013219.

[15] S. R. Sankhe and A. Ambhaikar, “Plant disease detection and classification techniques: A review,” Multiagent and Grid Systems, vol. 20, no. 3–4, pp. 265–282, 2025. doi: 10.1177/15741702241304087.

Authorized licensed use limited to: SRM Institute of Science and Technology Kattankulathur. Downloaded on March 22,2026 at 16:21:30 UTC from IEEE Xplore. Restrictions apply.